MILITARY SPECIFICATION

SHAMPOO, COAL TAR EXTRACT, SALICYLIC ACID, AND SULFUR LOTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Shampoo, Coal Tar Extract, Salicylic Acid and Sulfur Lotion (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186	- Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals. (See 6.3)

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a surface active, penetrating, therapeutic, antiseborrheic coal tar extract shampoo suitable for the relief of itching due to dandruff, seborrhea capitus and other seborrheic conditions.

3.1.1 Shall have the following formula:

Sulfur - - - - - - - 2.0 percent
Salicylic Acid - - - 2.0 percent
Water Miscible Coal Tar Extract, Approx. 1.33 percent
(Equivalent to 0.5 percent Coal Tar USP)
Colored, scented surface active base q.s. 100 percent.

3.2 Finished lotion.

3.2.1 Assay of ingredients in the finished lotion.

3.2.1.1 Coal tar extract. Shall contain not less than 0.45 percent and not more than 0.55 percent of Coal Tar, USP, when determined as specified in 4.7.1.

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3.2.1.2 Salicylic acid. Shall contain not less than 1.80 percent and not more than 2.20 percent when determined as specified in 4.7.2.

3.2.1.3 Sulfur. Shall contain not less than 1.80 percent and not more than 2.20 percent when determined as specified in 4.7.3.

3.2.2 pH. The undiluted lotion shall have a pH of not less than 3.45 and not more than 4.05 when determined potentiometrically at 25°C, using the USP method.

3.2.3 Surface tension. A 2 percent solution (w/w) in demineralized, distilled or purified water, after careful stirring to avoid excessive foaming, shall have an apparent surface tension of 32.1 to 33.1 dynes per centimeter, when determined between 20°C and 25°C, using a semi-automatic Fisher Surface Tensiomat, Cenco DuNouy Precision Tensiometer, or other suitable instrument, giving comparable results.

3.2.4 Loss on drying. Shall lose not less than 62.5 percent and not more than 67.5 percent of its initial weight when determined as specified in 4.7.4.

3.2.5 Ash. Shall have an ash content of not less than 5.7 percent and not more than 7.0 percent when determined as specified in 4.7.5.

3.2.6 Lather volume and permanence. The total volume of suds and water shall be not less than 65 ml initially, and not less than 60 ml after standing undisturbed for 30 minutes when determined as specified in 4.7.6.

3.3 Ingredients.

3.3.1 Coal Tar Extract.

3.3.1.1 Material. Shall be a concentrated solution of Coal Tar, USP in Polyoxyethylene - 4 - Lauryl Ether.

3.3.1.2 Assay. Shall contain not less than 25.0 percent and not more than 40.0 percent Coal Tar, USP, when determined as specified in 4.7.7.

3.3.2 Coal Tar, Salicylic Acid and Precipitated Sulfur. The Coal Tar, Salicylic Acid and Precipitated Sulfur used in the manufacture of the lotion shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3.3 Other ingredients. All other ingredients entering into the manufacture of the lotion shall be of USP or NF, quality, or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

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3.4 Delivery. Not more than 6 months shall have elapsed from the date of manufacture of the lotion to the date of delivery to the Government.

3.5 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of lotion produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

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4.3.1 For raw material and end item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay:			
Coal Tar Extract	3.2.1.1	S-2	5
Salicylic Acid	3.2.1.2	S-2	5
Sulfur	3.2.1.3	S-2	5
pH	3.2.2	S-2	5
Surface tension	3.2.3	S-2	5 Composite
Loss on drying	3.2.4	S-2	5
Ash	3.2.5	S-2	5
Lather volume and permanence	3.2.6	S-2	5
Ingredients:			
Coal Tar Extract	3.3.1	2/	-

1/ Where possible, the sample shall be used for two or more tests.

2/ Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.4 Examination. The finished solution shall be examined to determine compliance with all requirements of this specification. Nonconformances will be permitted to the extent indicated in 4.3.

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4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Lotion not free from foreign matter.
102	Lotion has crystal formation.
103	Lotion not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents and foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Coal tar. Accurately weigh approximately 1 gram of lotion into a tared 100 ml volumetric flask. Add about 50 ml Benzene, reagent grade, and shake well. Dilute to volume with the Benzene, mix well, and filter through No. 42 Whatman filter paper, preventing evaporation of the solvent during the procedure. Discard the first 25 ml of filtrate. Prepare a standard solution having a known final concentration of approximately 5 mg of Coal Tar, USP, in 100 ml of Benzene, reagent grade. Concomitantly determine the absorbances of the standard and sample solutions in a 1-cm cell at 428 nm, using a suitable spectrophotometer with Benzene, reagent grade, as the blank.

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Calculation:

$$\frac{Au \times Cs \times 10}{As \times SW} = \text{percent Coal Tar, USP, in lotion}$$

Where:

Au = Absorbance of the sample
As = absorbance of the standard
Cs = concentration of standard in grams per liter
SW = weight of sample in grams

4.7.2 Salicylic Acid.

Reagent: Dissolve 1.00 gram of $Fe(NO_3)_3$ and 1.0 ml concentrated HNO_3 in 100 ml purified water.

Standard solution: Accurately weigh approximately 100 mg of Salicylic Acid, USP Reference Standard into a 100 ml volumetric flask, dissolve in, and dilute to volume with SDA 3A alcohol.

Procedure: Accurately weigh about 200 mg of lotion into a tared 100 ml volumetric flask. Add 4.0 ml of SDA 3A alcohol and shake well.

Pipet 4.0 ml of the standard solution into a separate 100 ml flask. To standard and sample flasks, add approximately 50 ml of purified water, 5.0 ml of $Fe(NO_3)_3$ reagent, mix, dilute to volume with purified water, and mix well. Permit to stand at least 15 minutes filter solutions through #40 Whatman paper. (Solution must be clear). Determine the absorbances of sample and standard versus purified water, in a 1-cm cell, in a suitable spectrophotometer, at 525 nm.

Calculation:

$$\frac{Au \times Cs \times 4}{As \times \text{sample weight (mg)}} = \text{percent Salicylic Acid in the lotion}$$

Where:

Au = absorbance of sample
As = absorbance of standard
Cs = weight of standard in mg

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4.7.3 Sulfur

Standard solution: Prepare a solution of Precipitated Sulfur, USP, in Benzene having a known final concentration of approximately 0.2 mg per ml.

Procedure: Accurately weigh approximately 100 mg of lotion into a beaker, disperse in 20 ml of purified water, and transfer quantitatively to a 60 ml separator using small portions of purified water. Pipet 10.0 ml Methylene Chloride, reagent grade to the separator, stopper and shake well. Permit the layers to separate, and run off approximately 5 ml of the lower layer (organic solvent) through phase separating paper such as Whatman 1PS, or equivalent. Collect the clear filtrate in an Erlenmeyer flask and stopper to prevent evaporation.

Using a prepared thin layer chromatography plate containing Silica Gel GF, with a uniform layer of thickness of approximately 250 microns, separately spot 10 µl of standard solution and 10 µl of clear filtrate sample solution approximately 4.0 cm from the bottom of the plate. Develop to a height of approximately 10 cm using Benzene as the mobile solvent in a solvent chamber previously saturated with Benzene vapor.

Permit the plates to dry thoroughly in air at room temperature, and quantitate the spots corresponding to sulfur (Rf approx. 0.7) using a recording densitometer, with exciting light at 254 nm and emitted light measured at 520 nm. Set the instrument for optimal operating conditions. Integrate the areas under the curve for standard and sample.

Calculation:

$$\frac{Qu \times Cs \; 1000}{Qs \times \text{Weight Sample (mg)}} = \text{percent sulfur in lotion}$$

Where:

Qu = Quantitation value sample
Qs = Quantitation value standard
Cs = Concentration standard mg per ml

4.7.4 Loss on drying. Accurately weigh approximately 6.5 grams of lotion into a tared aluminum weighing dish and dry in an oven at 110°C. ± 2°C for 4 hours.

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4.7.5 Ash. Accurately weigh approximately 10 grams of sample into a tared crucible and dry overnight at approximately 110°C. Place in a muffle oven and slowly permit the temperature to reach 800°C + 25°C. Maintain at this temperature for 15 to 20 minutes. Remove, cool, and calculate the percent ash.

4.7.6 Lather volume and permanence. Accurately weigh 65 mg of lotion into a 100 ml graduated glass stoppered cylinder. Add 20 ml of purified water and shake vigorously for 30 seconds.

4.7.7 Coal tar extract (assay of raw material). Accurately weigh about 130 mg of sample into a tared beaker. Dissolve in, and quantitatively transfer to a 100 ml volumetric flask using Benzene, reagent grade. Mix well and dilute to volume with the Benzene. Pipet 10.0 ml of the solution into another 100 ml volumetric flask and dilute to volume with the Benzene.

Prepare a standard solution of Coal Tar, USP, having a known final concentration of approximately 50 mg per liter in Benzene, reagent grade. Concomitantly determine the abosrbances of the standard and sample solutions in 1 cm cell at 428 nm, using a suitable spectrophotometer with Benzene, reagent grade, as the blank.

Calculation:

$$\frac{Au \times Cs \times 100}{As \times Sw} = \text{percent coal tar in the extract.}$$

Where:

Au = absorbance of the sample
As = absorbance of the standard
Cs = concentration of standard in grams per liter
Sw = weight of sample in grams

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (See 6.3), and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (See 6.1).

5.2.1 Unit of issue. One bottle containing 4 oz (118 ml), as specified, constitutes one unit of issue.

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5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit packaging. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate packaging. Thirty-six (36) units of issue shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the date of manufacture prefixed by "MFD".

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5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in paragraph 5.5.3 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed.

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock Number	Item Identification
6508-00-116-1367	SHAMPOO, COAL TAR EXTRACT, SALICYLIC ACID, AND SULFUR LOTION, Medicated, 4 **fl oz (118 ml)**

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6508-0001

☆ U. S. GOVERNMENT PRINTING OFFICE: 1976-703-020/286

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-37426 SHAMPOO, COAL TAR EXTRACT, SALICYLIC ACID, AND SULFUR LOTION

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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